

INTRODUCTION BROCHURE

OpenSourceRail

An open stack for affordable, locally buildable urban rail.

OpenSourceRail combines city design automation, a simulator and control stack, parametric mechanical CAD, hardware reference designs, operations documentation, and a machine-checkable safety case.

GoA 4

Driverless reference operation.

51 m

Reference 3-car light metro trainset.

build123d

CAD generated from source, not hand-edited STEP files.

Software: Apache 2.0. Hardware designs: CERN-OHL-S v2. Documentation: CC-BY-SA 4.0.



1. The Problem

Urban rail is too slow and expensive to deploy

- Proprietary vehicle and signalling packages create lock-in.
- Capital cost makes medium-size corridors hard to justify.
- Integration evidence is often document-heavy and hard to regenerate.
- Local manufacturers are pushed toward low-value installation work.

OpenSourceRail's bet

- Publish reference designs and machine-readable interfaces.
- Use commodity hardware and certified COTS modules where sensible.
- Keep the safety case explicit, testable, and versioned with the design.
- Make local fabrication the default: cut, weld, bond, bolt, test.

Open

Readable source artifacts.

Repeatable

Regenerate cities, CAD, BOMs, docs.

Modular

Swap suppliers if envelopes fit.

Assessable

Evidence links to tests and drawings.

2. System Overview

Design automation

GIS and city templates generate alignments, station placements, maps, cost summaries, and simulator scenarios.

Control software

Rust crates cover simulator, interlocking, ATP, brake, obstacle detection, TCMS, GUIs, and safety-case compilation.

Mechanical package

build123d source generates trainsets, bogies, track, civil works, stations, depots, and STEP handoff files.

Hardware hosts

Reference SBC/ECU classes define train, wayside, station, and obstacle-detection hosts.

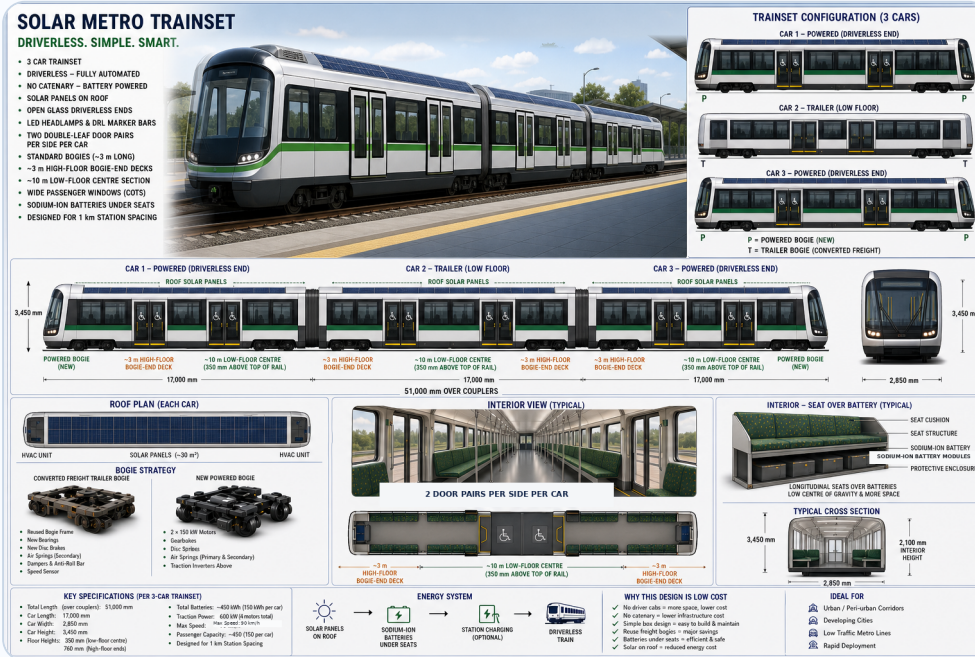
Operations docs

Rulebooks, staff procedures, maintenance flows, and depot processes are part of the same repository.

Safety evidence

Certification and GSN documents capture assumptions, tests, and remaining evidence gaps.

3. Reference Vehicle



Solar metro trainset

- 3-car, 51 m reference consist.
- 17 m repeated car module.
- Two low-floor door pairs per side per car.
- About 10 m low-floor centre section.
- About 3 m high-floor end decks over standard bogies.
- Open glass driverless ends at both train ends.
- Under-seat sodium-ion batteries and roof solar.

Design authority

build123d source in mechanical-py/src/osr_mech/rolling_stock; PNG and STEP files are generated handoff artifacts.

4. Trainset Layout



Why 17 m cars?

The 17 m module keeps a 3-car train at 51 m, leaves platform margin on a 67 m standard station, and remains workshop-friendly.

- Standard bogies sit at each car end.
- Raised decks cover bogie structure and service access.
- The low-floor middle supports doors, PRM bays, and passenger flow.
- Capacity scales by adding repeated cars or by a later 19-20 m variant.

2 + 4

Two powered bogies, four trailer bogies.

12

Exterior door cassettes per consist.

450 kWh

Reference sodium-ion onboard energy.

5. Production Assembly



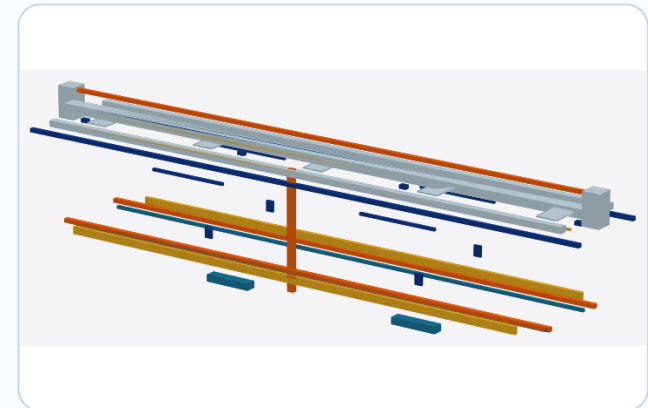
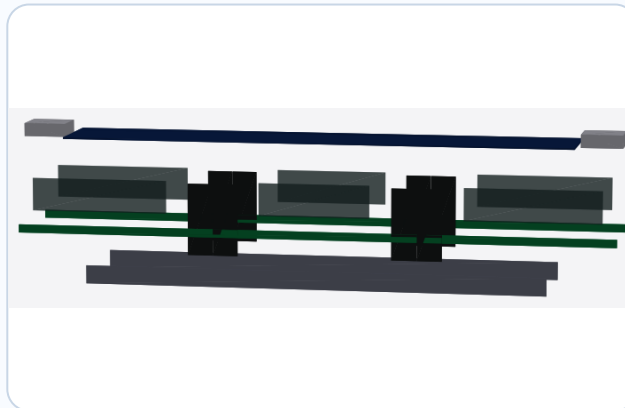
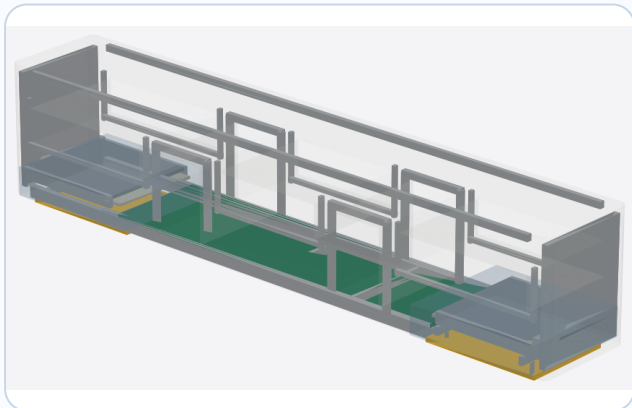
Make the steel datum structure locally; install supplier modules late; repeat the 17 m car body; lower onto standard bogies.

6. Layered Car Body

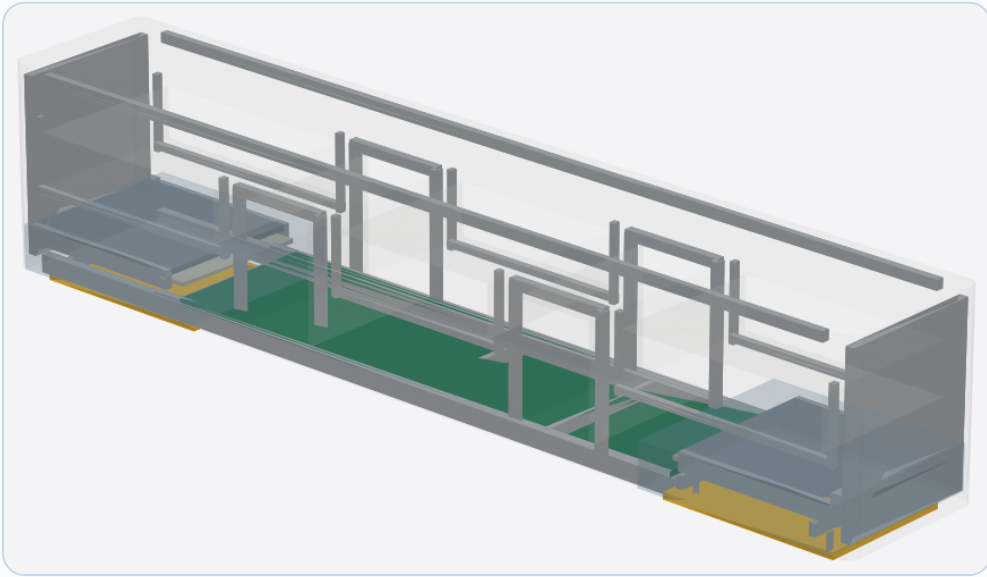


Components to final assembly

- Primary steel shell, floor, and portal structure.
- Exterior cladding, glazing, doors, livery, solar, and skirts.
- Interior fit-out with seats, PRM bays, grab rails, and displays.
- Service layers for HVAC, LV/data, HV traction, PV, coolant, and fire paths.



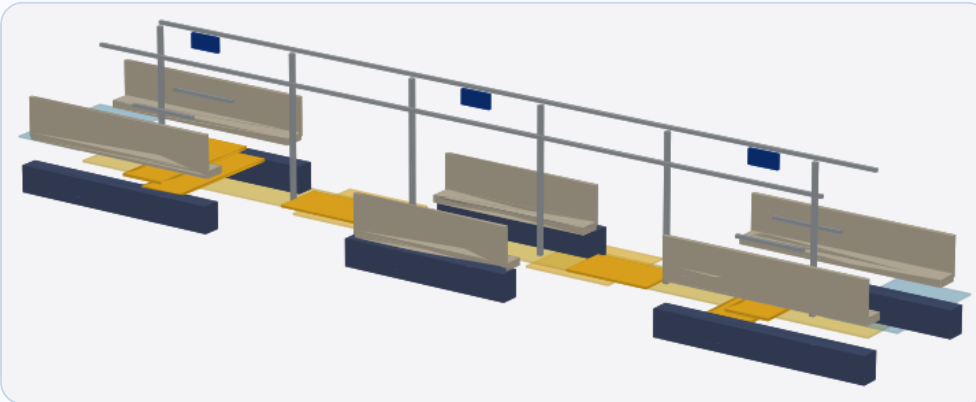
7. Body Structure And Load Path



Fabricated, inspectable steel

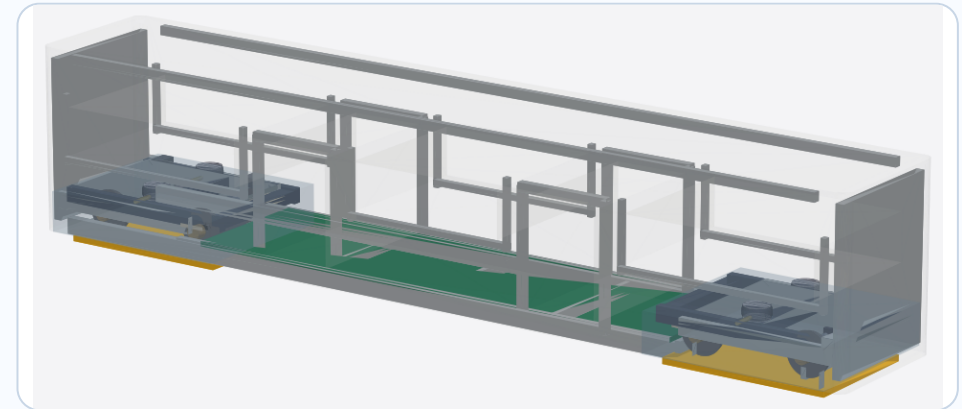
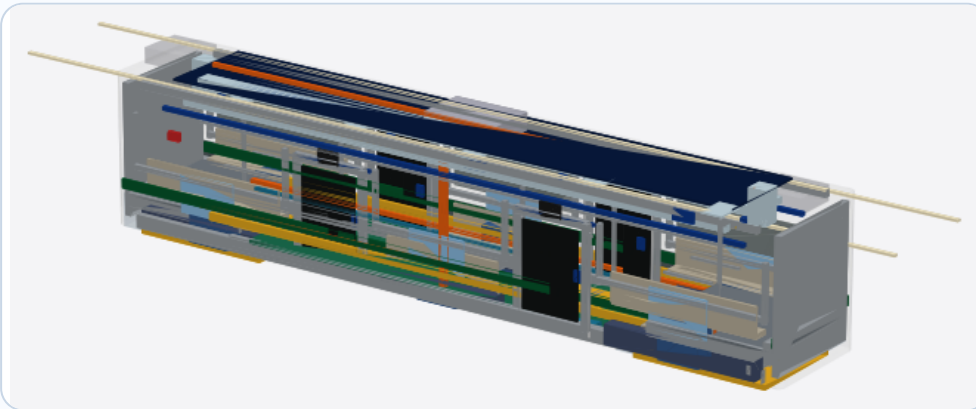
- S355 underframe ladder and side/roof spaceframe.
- Door portal posts and headers around COTS cassettes.
- Window posts, waist rails, cantrails, and end rings.
- Anti-climber beam and open-glass cowl structural ring.
- Bogie rotation, suspension-travel, and drop-zone envelopes.

8. Access And Interior



Passenger design intent

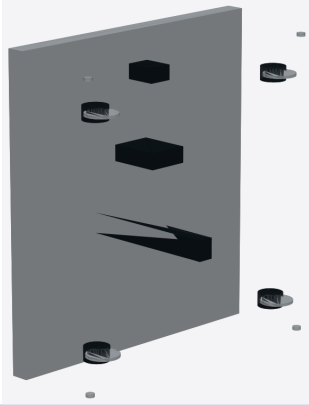
- Two door pairs per side per car, concentrated in the low-floor zone.
- Wheelchair turning circles and PRM bays in the centre section.
- Longitudinal seats over under-seat battery strakes.
- Open passenger view through both driverless glass ends.
- Handrails and step treads where the floor rises over bogies.



9. Driverless Ends And Sensing

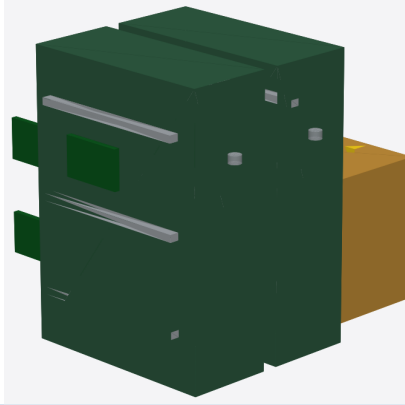
Open glass ends

The ends are not cab partitions. Passengers retain a forward/rear view through heated RF-transparent glass.



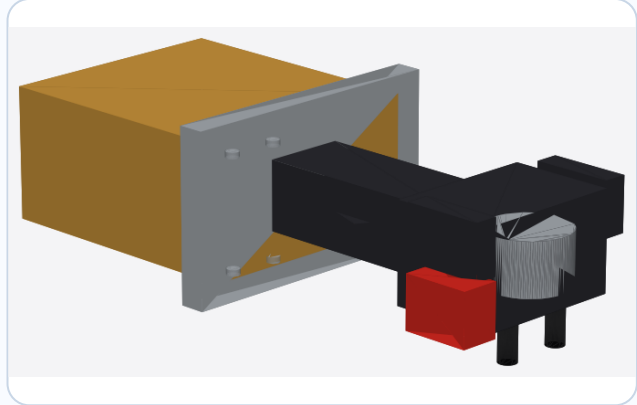
T-OBS sensor pack

LIDAR, mmWave radar, stereo cameras, and ultrasonic sensors sit behind serviceable cowl hardware.

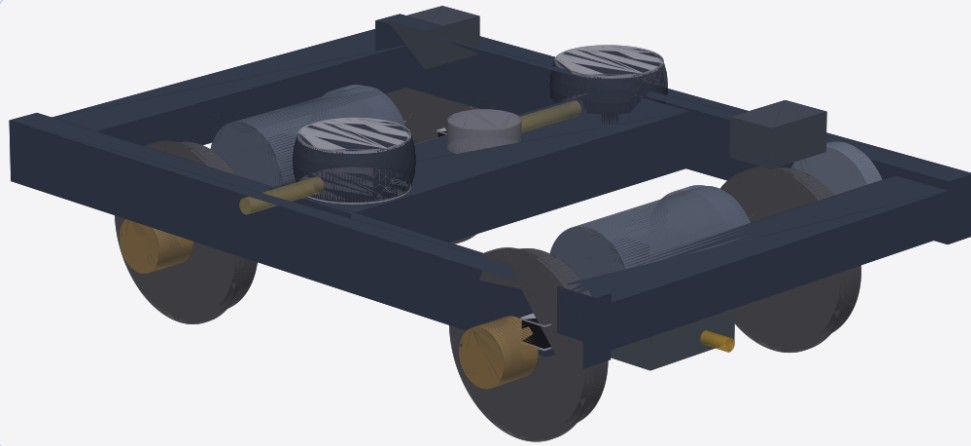


Recovery only

Hidden maintenance/recovery controls are not a normal driver cab.

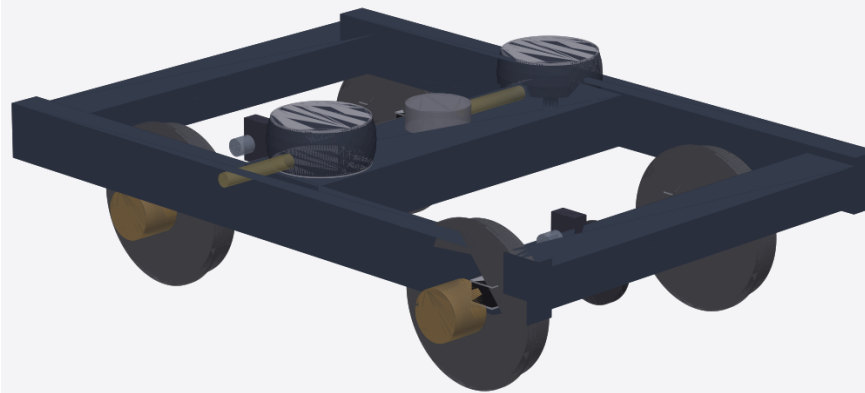


10. Bogies And Running Gear

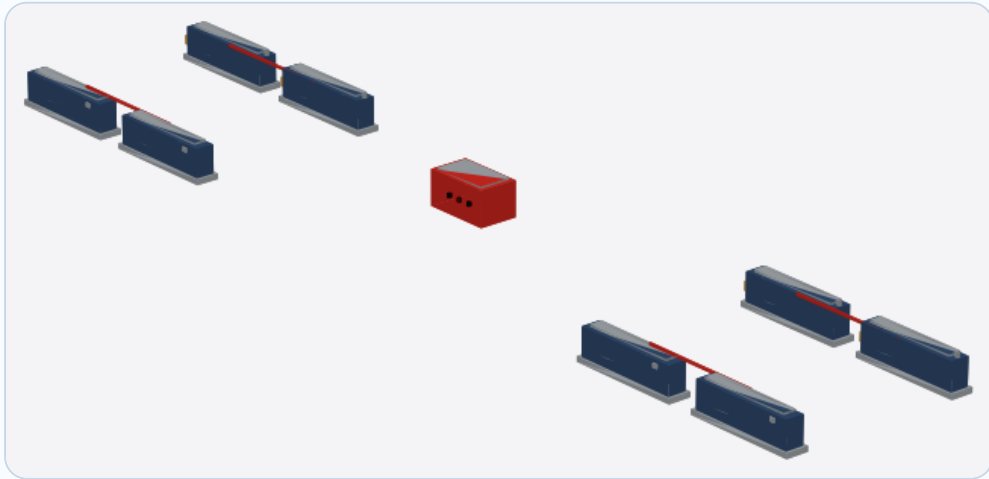


Standardized bogie strategy

- Two powered bogies total, at the outer end cars.
- Four trailer bogies using the common frame and suspension envelope.
- Powered bogie includes PMSM motors, gearboxes, brakes, and suspension.
- Trailer bogie keeps the same maintainable running-gear architecture without traction drives.

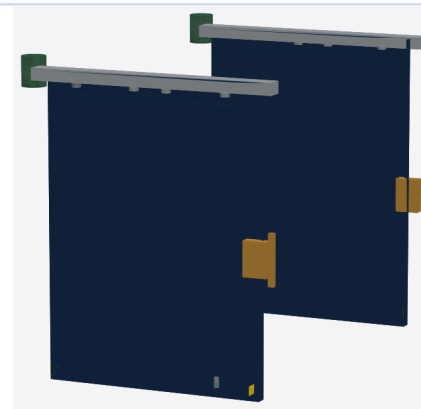
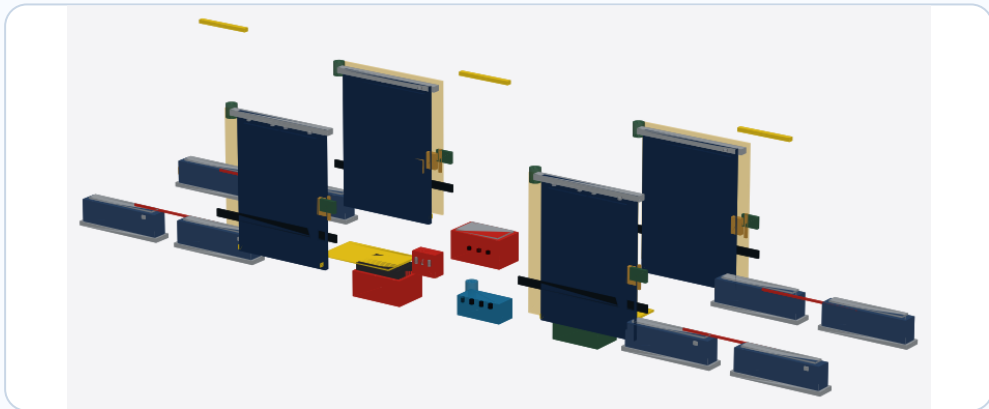


11. Energy System

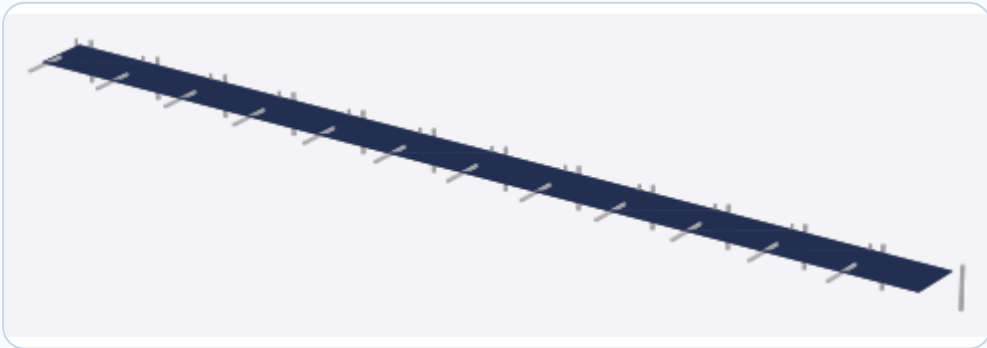


Battery electric, catenary-free

- Reference 450 kWh sodium-ion consist energy.
- Under-seat modules keep the roof available for solar and HVAC.
- Station side-pin charging is the primary concept.
- Roof PV supports auxiliary energy and site energy modelling.



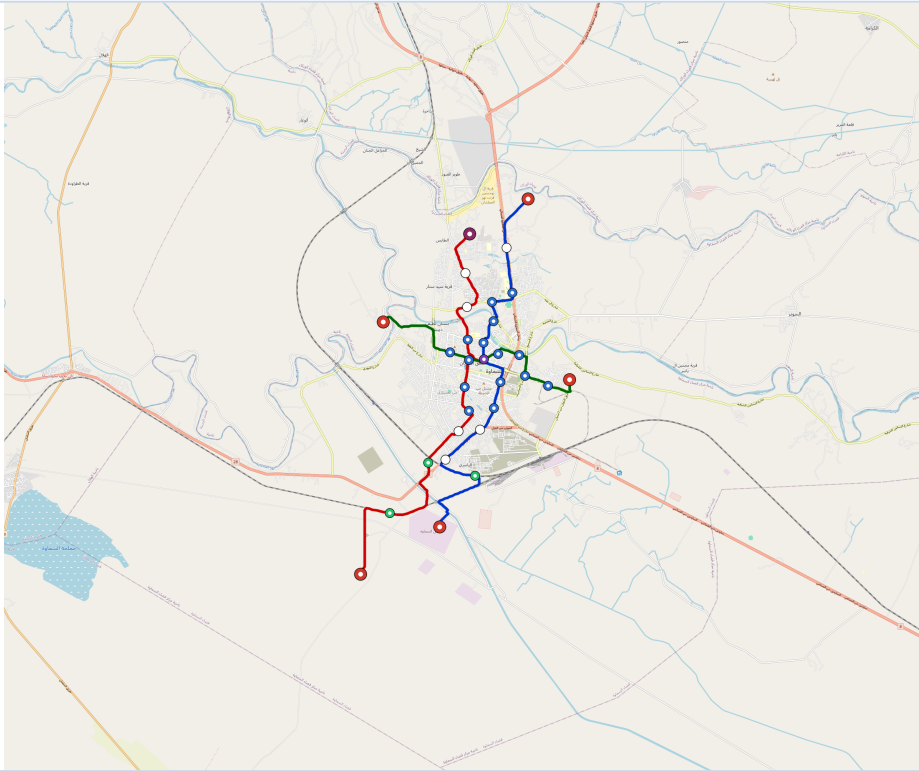
12. Stations And Civil Works



Station pattern

- Low platform interface at the 350 mm train floor datum.
- Platform screen-door alignment and train-side interlocks are reserved in CAD.
- Station charging can be side-pin or pantograph-down depending on civil geometry.
- Stations, track, and city designs regenerate from machine-readable templates.

13. City Design Automation



From city inputs to runnable scenarios

- World sample batches define candidate cities.
- GIS tooling emits route GeoJSON, maps, quality YAML, and README summaries.
- Rust design synthesis produces simulator scenarios and cost summaries.
- Every city folder is a reproducible design package.

14. Simulator And Control Stack



Software crates

- Simulator and scenario execution.
- Interlocking, ATP, brake, and obstacle-detection crates.
- TCMS and GUI tooling.
- Safety-case compiler and verification support.



15. Hardware Host Classes

T-ECU/S

Train safety kernel host for safety-related train functions.

T-ECU/A

Train application computer for non-safety application logic and integrations.

T-OBS

Obstacle-detection host connected to the driverless nose sensor pack.

W-SBC

Wayside controller class for station and trackside equipment.

S-SBC

Station/depot host for local operations and maintenance integration.

DIY pilots

Commodity-module assembly path for lab and early integration builds.

16. Safety And Certification

Safety case is part of the product

The repository includes RFCs, safety-case documents, certification packs, operations rules, and formal models. The project does not certify deployments; it produces artifacts suitable for independent assessment.

Evidence direction

- Requirements and assumptions are explicit.
- Tests and generated artifacts are linked to source.
- Supplier documents fit controlled interface envelopes.
- Authority review remains deployment-specific.

Concept

Reference architecture and RFCs.

v1

Envelope CAD, BOM, docs, simulation.

v2

Supplier envelopes and shop drawings.

First article

Build, inspect, static test.

Assessment

Independent review and authority process.

17. Manufacturing Doctrine

Scope	Default	Reason
Primary structure	MAKE: cut, bend, weld S355 steel	Available in rail, truck, ship, oilfield, and bridge workshops.
Panels	BID/SOURCE composite side, roof, and cowl panels	Low tooling cost and replaceable after damage.
Doors/windows/HVAC	COTS or BID supplier modules	Use certified rail/bus module evidence instead of bespoke development.
Bogies	Qualified MAKE plus supplier-certified rotating/brake parts	Structural frame can be local; wheelsets and brakes need supplier evidence.
Traction and BMS	BID/SOURCE packages	Supplier qualification, EMC, and lifecycle test evidence required.

18. Cost And BOM Direction

\$2.21M

Direct material planning estimate for 3-car consist.

\$2.98M

Planning-grade consist cost after shop labor uplift.

12

COTS electric door cassettes per consist.

18

COTS window cassettes per consist.

MAKE

Fabricator-owned steel datum parts, brackets, harness supports, and local tooling.

BID/SOURCE

Doors, windows, HVAC, electronics, traction, bogie parts, and certified passenger modules.

19. Operations And Maintenance

Operations

Rulebook, OCC guidance, station staff procedures, and depot flows are kept with the technical design.

Maintenance

Design emphasizes bolted service cassettes, removable skirts, access lids, and supplier module replacement.

Recovery

Driverless operation includes rescue couplers, recovery cabinets, and explicit degraded-mode procedures.

Local skills

Welding, bonding, wiring, inspection, and commissioning cells.

Supplier evidence

Installation, maintenance, lifecycle, and certification documents remain required.

20. Deployment Workflow

1. Select corridor

Use city generation and GIS evidence.

2. Configure system

Pick consist, station, depot, energy, and operations templates.

3. Freeze envelopes

Supplier modules must fit reserved interfaces.

4. Build first article

Fabricate, install COTS modules, inspect.

5. Certify locally

Submit evidence to national authority and independent assessor.

Regenerable artifacts

City designs, BOM CSVs, CAD screenshots, STEP catalogues, test outputs, and document books can be regenerated from repository source.

21. Repository Map

Path	Purpose
crates/	Rust simulator, control, safety, GUI, and design tools.
design-py/ and designs/	GIS sidecar and generated city design packages.
mechanical-py/	build123d mechanical source and generated STEP catalogue.
hardware/	Hardware reference designs and host-class integration.
docs/	Architecture, RFCs, operations, certification, safety case, rolling-stock docs.
lib/	Templates, recipes, cost inputs, city batches, and examples.
formal/	TLA+ consensus specification and model-checking harnesses.

22. Regeneration Commands

Repository health

```
python3 scripts/repo-health.py --quiet
```

Mechanical tests

```
PYTHONPATH=mechanical-py/src pytest mechanical-py/  
tests -q
```

CAD screenshots

```
PYTHONPATH=mechanical-py/src python3 mechanical-py/  
scripts/render_screenshots.py
```

STEP catalogue

```
PYTHONPATH=mechanical-py/src python3 -m  
osr_mech.catalog --out mechanical-py/catalog
```

BOM CSV

```
python3 scripts/export-light-metro-bom.py
```

City regeneration

```
scripts/regenerate-city.sh samawah
```

23. v0.2 Focus

Rolling stock

- Supplier exact envelopes for doors, windows, HVAC, traction, and bogie components.
- 2D shop drawings, weld maps, tolerance stacks, and inspection plans.
- Harness clamp locations and maintainability clearances.
- FEA-ready brackets and controlled drawing register.

System integration

- Station charging interface and platform screen-door proof.
- Hardware bring-up and simulated end-to-end control paths.
- Expanded certification evidence and test traceability.
- More generated cities and cost comparisons.

24. What To Read Next

Core entry points

- README.md for the repository overview.
- docs/ARCHITECTURE.md for system structure.
- docs/rolling-stock/light-metro-3car/README.md for the reference train.
- docs/rfcs/README.md for design decisions.
- docs/certification/ and docs/safety-case/ for assessment evidence.

Open artifact, not a certifier

OpenSourceRail produces open designs, software, hardware references, and evidence packs. Deployment partners and national authorities remain responsible for supplier qualification, safety assessment, homologation, and operation.

Start with the generated trainset and production graphics, then follow the build123d source.